SOLUTIONS:

Read the passage below then answer the questions which follow.

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- A solution is a special kind of mixture. A solution is formed when one chemical dissolves, or mixes evenly into another. The most common type of solution is formed when a solid, such as copper sulphate, dissolves in a liquid as water. A copper sulphate solution is formed.

- In a solution no solid particles can be seen. They have all mixed evenly into the liquid. However, we can usually tell that the solid has dissolved. A copper sulphate solution, for example is blue. The colour due to the dissolved copper sulphate crystals is obvious. A sodium chloride solution looks just like water. However, if we taste it we know that salt has been dissolved in it because of the salty flavour.

- We call a liquid (like water) which will dissolve another chemical (like salt) a solvent for that chemical. The salt, or any other solvent for sand because sand will not dissolve in water.

- Copper sulphate and salt are both solutes in water. That is, they both dissolve in water. However, neither of them are solutes for a liquid like methylated spirits because they do not dissolve in it.

- We say that copper sulphate and salt are soluble in water, but insoluble in methylated spirits.

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Questions

1. Use the information in the passage to write a definition of the following
   (a) dissolve

   (b) insoluble

   (c) soluble

   (d) solute

   (e) solution

   (d) solvent

2. The passage on the previous page contains 5 paragraphs of information. Each paragraph contains a **main idea** as well as an example. Using one sentence only for each paragraph describe the main ideas in each paragraph.

3. For the following sentences write whether they are **TRUE or FALSE.**
   (a) Copper sulphate is a blue chemical and it is **soluble** in water.
   (b) Sodium chloride does not **dissolve** in water.
   (c) Water is not a **solvent** for sand.
   (d) Salt is insoluble in methylated spirits.
   (e) When Copper sulphate dissolves in water it forms a **solution.**

4. The following are all solutions each contains a solute dissolved in a solvent. For each solution, name the **solute** and the **solvent.**
   (a) sea water
   (b) swimming pool water
   (c) a cup of coffee
   (d) a bowl of jelly